JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2025 29th January Morning Shift

Question Paper with Answer Key

75	ENGLISH	DARK
QUESTIONS	LANGUAGE	FORMAT

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

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1.24 g of AX_2 (molar mass 124 g mol^{-1}) is dissolved in 1 kg of water to form a solution with boiling point of 100.0156°C, while 25.4 g of AY_2 (molar mass 250 g mol^{-1}) in 2 kg of water constitutes a solution with a boiling point of 100.0260°C.

$K_b(H_2O)$ = 0.52 kg mol^{-1}

Which of the following is correct?

(A) AX_2 and AY_2 (both) are completely unionised.

(B) AX_2 and AY_2 (both) are fully ionised.

(C) AX_2 is completely unionised while AY_2 is fully ionised.

(D) AX_2 is fully ionised while AY_2 is completely unionised.

Q2. MCQ (Single Correct Answer)

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For a Mg | Mg^{2+} (aq) || Ag^{+} (aq) | Ag the correct Nernst Equation is :

(A) $E_{cell} = E^{o}_{cell} - \frac{RT}{2F}\ln\frac{[Ag^{+}]^2}{[Mg^{2+}]}$

(B) $E_{cell} = E^{o}_{cell} - \frac{RT}{2F}\ln\frac{[Mg^{2+}]}{[Ag^{+}]}$

(C) $E_{cell} = E^{o}_{cell} - \frac{RT}{2F}\ln\frac{[Ag^{+}]}{[Mg^{2+}]}$

(D) $E_{cell} = E^{o}_{cell} + \frac{RT}{2F}\ln\frac{[Ag^{+}]^2}{[Mg^{2+}]}$

Q3. MCQ (Single Correct Answer)

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Match List - I with List - II.

List - I (Carbohydrate)	List - II (Linkage Source)
(A) Amylose	(I) β - $C_1 - C_4$, plant
(B) Cellulose	(II) α - $C_1 - C_4$, animal
(C) Glycogen	(III) α - $C_1 - C_4$, α - $C_1 - C_6$, plant
(D) Amylopectin	(IV) α - $C_1 - C_4$, plant

Choose the correct answer from the options given below :

(A) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(B) (A)-(III), (B)-(III), (C)-(I), (D)-(IV)

(C) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(D) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Q4. MCQ (Single Correct Answer)

The correct increasing order of stability of the complexes based on Δ_0 value is :

I. I. $[Mn(CN)_6]^{3-}$

II. II. $[Co(CN)_6]^{4-}$

III. III. $[Fe(CN)_6]^{4-}$

IV. IV. $[Fe(CN)_6]^{3-}$

(A) I < II < IV < III

(B) IV < III < II < I

(C) III < II < IV < I

(D) II < III < I < IV

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Q5. MCQ (Single Correct Answer)

500 J of energy is transferred as heat to 0.5 mol of Argon gas at 298 K and 1.00 atm. The final temperature and the change in internal energy respectively are: Given: R = 8.3 J K^{-1} mol^{-1}

(A) 368 K and 500 J (B) 348 K and 300 J (C) 378 K and 300 J (D) 378 K and 500 J

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Q6. MCQ (Single Correct Answer)

The molar conductivity of a weak electrolyte when plotted against the square root of its concentration, which of the following is expected to be observed?

(A) Molar conductivity increases sharply with increase in concentration.

(B) Molar conductivity decreases sharply with increase in concentration.

(C) A small increase in molar conductivity is observed at infinite dilution.

(D) A small decrease in molar conductivity is observed at infinite dilution.

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Q7. MCQ (Single Correct Answer)

The correct option with order of melting points of the pairs (Mn, Fe), (Tc, Ru) and (Re, Os) is :

(A) Fe < Mn, Ru < Tc and Re < Os

(B) Fe < Mn, Ru < Tc and Os < Re

(C) Mn < Fe, Tc < Ru and Re < Os

(D) Mn < Fe, Tc < Ru and Os < Re

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Q8. MCQ (Single Correct Answer)

The reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism:

$A_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A + A$ (fast)

$A + B_2 \xrightarrow{k_2} AB + B$ (slow)

$A + B \rightarrow AB$ (fast)

The overall order of the reaction is:

(A) 3 (B) 2.5 (C) 1.5 (D) 2

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Q9. MCQ (Single Correct Answer)

If a_0 is denoted as the Bohr radius of hydrogen atom, then what is the de-Broglie wavelength (λ) of the electron present in the second orbit of hydrogen atom? [n : any integer]

(A) $\frac{8\pi a_0}{n}$ (B) $\frac{2a_0}{n\pi}$ (C) $\frac{na_0}{4\pi}$ (D) $\frac{4\pi a_0}{n}$

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Q10. MCQ (Single Correct Answer)

The standard reduction potential values of some of the p-block ions are given below. Predict the one with the strongest oxidising capacity.

(A) $E^o_{Sn^{4+}/Sn^{2+}} = +1.15\ V$ (B) $E^o_{Al^{3+}/Al} = -1.66\ V$

(C) $E^o_{Pb^{4+}/Pb^{2+}} = +1.67\ V$ (D) $E^o_{Tl^{3+}/Tl} = +1.26\ V$

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Q11. MCQ (Single Correct Answer)

Choose the **correct** statements.

(A) Weight of a substance is the amount of matter present in it.

(B) Mass is the force exerted by gravity on an object.

(C) Volume is the amount of space occupied by a substance.

(D) Temperatures below 0°C are possible in Celsius scale, but in Kelvin scale negative temperature is not possible.

(E) Precision refers to the closeness of various measurements for the same quantity.

Choose the **correct** answer from the options given below :

(A) (B), (C) and (D) Only (B) (A), (B) and (C) Only

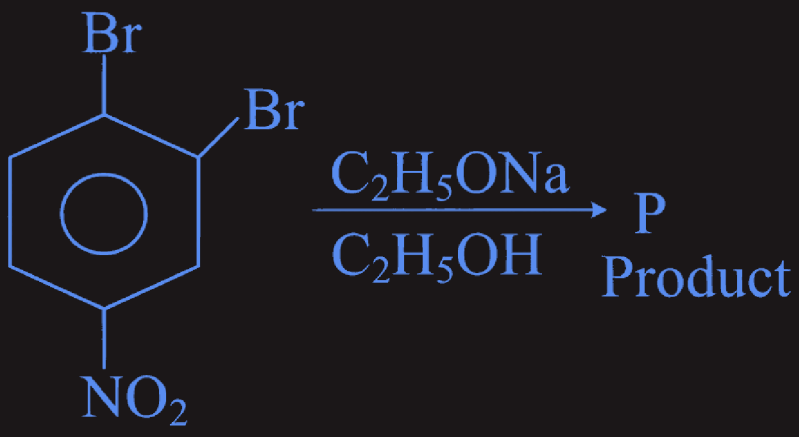
(C) (C), (D) and (E) Only (D) (A), (D) and (E) Only

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Q12. MCQ (Single Correct Answer)

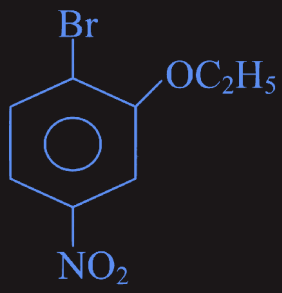
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In the following substitution reaction :

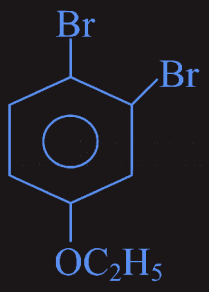


Product 'P' formed is :

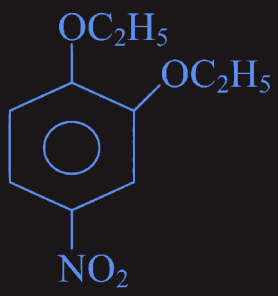
(A)



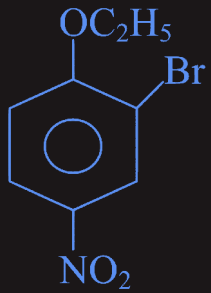
(B)



(C)



(D)



Q13. MCQ (Single Correct Answer)

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At temperature T, compound $AB_{2(g)}$ dissociates as $AB_{2(g)} \rightleftharpoons AB_{(g)} + \frac{1}{2}B_{2(g)}$ having degree of dissociation x (small compared to unity). The correct expression for x in terms of K_p and p is:

(A) $\sqrt{K_p}$

(B) $\sqrt[3]{\frac{2K_p^2}{p}}$

(C) $\sqrt[3]{\frac{2K_p}{p}}$

(D) $\sqrt[4]{\frac{2K_p}{p}}$

Q14. MCQ (Single Correct Answer)

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Match List - I with List - II.

List - I (Complex)	List - II (Hybridisation & Magnetic characters)
(A) $[MnBr_4]^{2-}$	(I) d^2sp^3 & diamagnetic
(B) $[FeF_6]^{3-}$	(II) sp^3d^2 & paramagnetic
(C) $[Co(C_2O_4)_3]^{3-}$	(III) sp^3 & diamagnetic
(D) $[Ni(CO)_4]$	(IV) sp^3 & paramagnetic

Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(I), (C)-(I), (D)-(IV)

(B) (A)-(III), (B)-(II), (C)-(III), (D)-(IV)

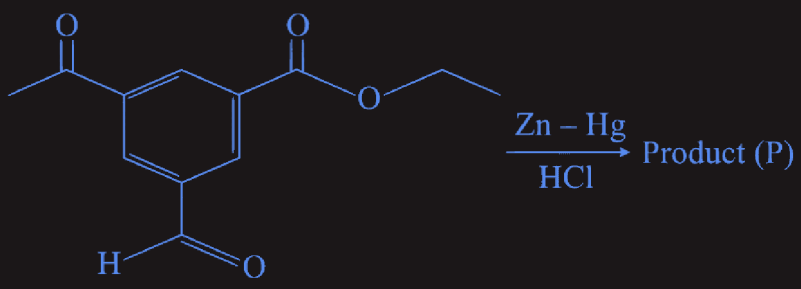
(C) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(D) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

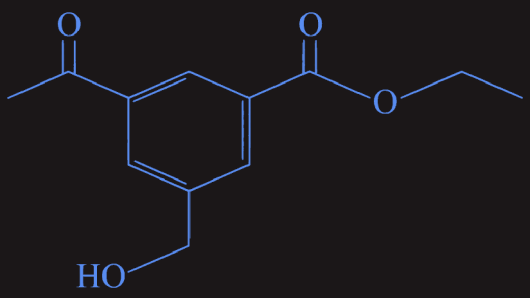
Q15. MCQ (Single Correct Answer)

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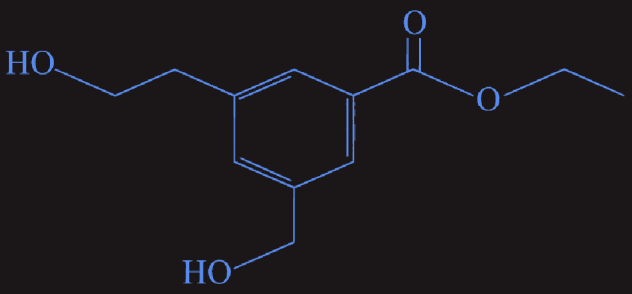
The product (P) formed in the following reaction is :



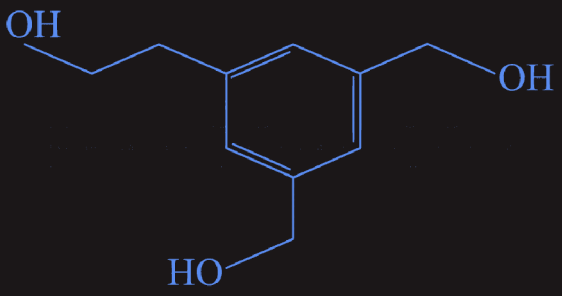
(A)



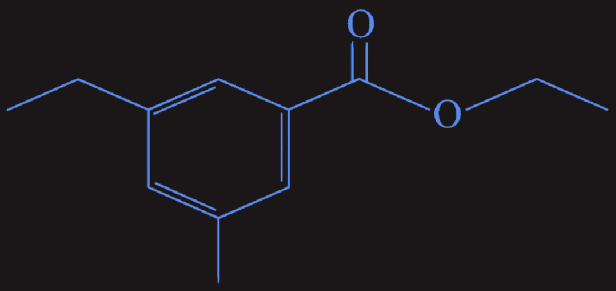
(B)



(C)



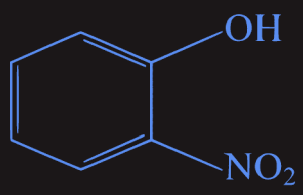
(D)

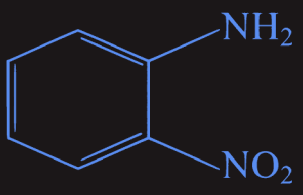


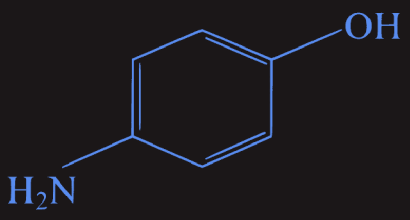
Q16. MCQ (Single Correct Answer)

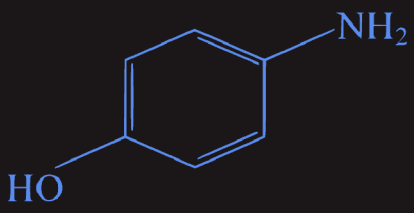
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The steam volatile compounds among the following are :

(A) 

(B) 

(C) 

(D) 

Choose the correct answer from the options given below :

(A) (B) and (D) Only

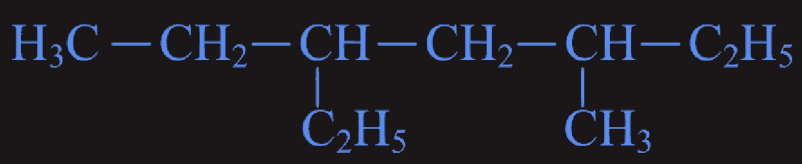
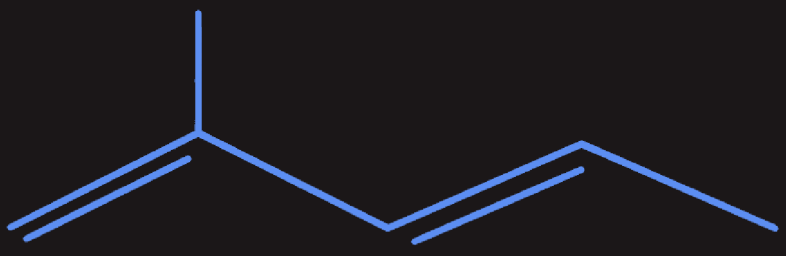
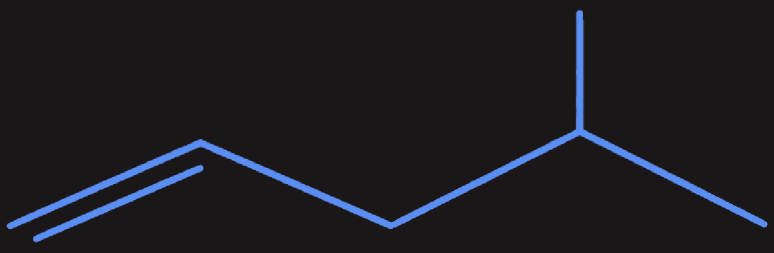
(B) (A), (B) and (C) Only

(C) (A) and (B) Only

(D) (A) and (C) Only

Q17. MCQ (Single Correct Answer)

Match List - I with List - II.

List - I (Structure)	List - II (IUPAC Name)
(A) 	(I) 4-Methylpent-1-ene
(B) $(CH_3)_2C(C_3H_7)_2$	(II) 3-Ethyl-5-methylheptane
(C) 	(III) 4,4-Dimethylheptane
(D) 	(IV) 2-Methyl-1,3-pentadiene

Choose the correct answer from the options given below :

(A) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

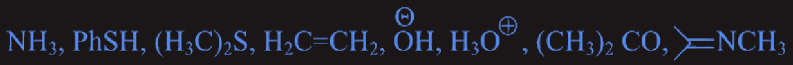
(B) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(C) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(D) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Q18. MCQ (Single Correct Answer)

Total number of nucleophiles from the following is :



(A) 4

(B) 6

(C) 5

(D) 7

Q19. MCQ (Single Correct Answer)

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An element 'E' has the ionisation enthalpy value of 374 kJ mol^{-1}. 'E' reacts with elements A, B, C and D with electron gain enthalpy values of −328, −349, −325 and −295 kJ mol^{-1}, respectively. The correct order of the products EA, EB, EC and ED in terms of ionic character is :

(A) EB > EA > EC > ED

(B) EA > EB > EC > ED

(C) ED > EC > EA > EB

(D) ED > EC > EB > EA

Q20. MCQ (Single Correct Answer)

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Given below are two statements :

Statement (I) : The radii of isoelectronic species increases in the order.

$Mg^{2+} < Na^{+} < F^{-} < O^{2-}$

Statement (II) : The magnitude of electron gain enthalpy of halogen decreases in the order.

Cl > F > Br > I

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Statement I is correct but Statement II is incorrect

(D) Both Statement I and Statement II are incorrect

Q21. Numerical

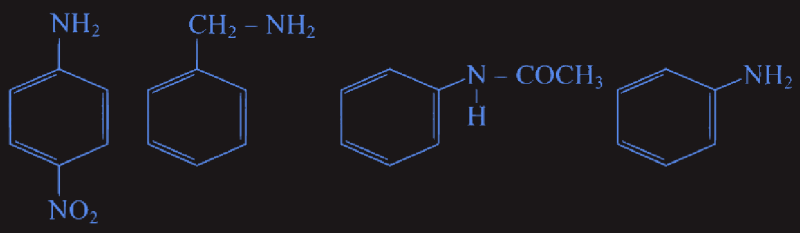
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The molar mass of the water insoluble product formed from the fusion of chromite ore ($FeCr_2O_4$) with Na_2CO_3 in presence of O_2 is _____ g mol^{-1}.

Q22. Numerical

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Given below are some nitrogen containing compounds :



Each of them is treated with HCl separately. 1.0 g of the most basic compound will consume _____ mg of HCl.

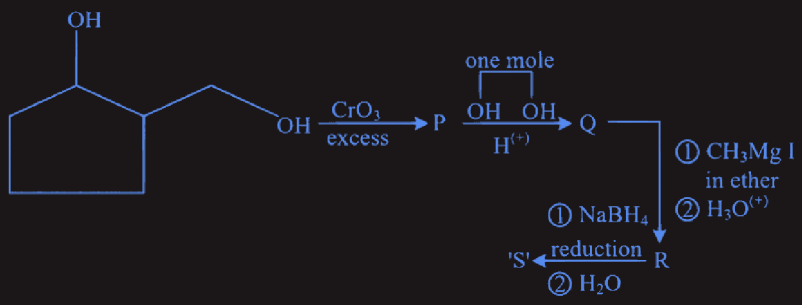
(Given molar mass in g mol^{-1} C : 12, H : 1, O : 16, Cl : 35.5)

Q23. Numerical

If A_2B is 30% ionised in an aqueous solution, then the value of van't Hoff factor (i) is _____ $\times 10^{-1}$.

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Q24. Numerical



0.1 mole of compound 'S' will weigh _____ g.

(Given molar mass in g mol^{-1} C : 12, H : 1, O : 16)

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Q25. Numerical

The sum of sigma (σ) and pi (π) bonds in Hex-1,3-dien-5-yne is _____.

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Let $L_1 : \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-1}{2}$ and $L_2 : \frac{x+1}{-1} = \frac{y-2}{2} = \frac{z}{1}$ be two lines.

Let L_3 be a line passing through the point (α, β, γ) and be perpendicular to both L_1 and L_2. If L_3 intersects L_1, then $|5\alpha - 11\beta - 8\gamma|$ equals :

(A) 25 (B) 20 (C) 16 (D) 18

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Q27. MCQ (Single Correct Answer)

Let P be the set of seven digit numbers with sum of their digits equal to 11. If the numbers in P are formed by using the digits 1, 2 and 3 only, then the number of elements in the set P is :

(A) 164 (B) 158 (C) 161 (D) 173

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Q28. MCQ (Single Correct Answer)

Let $A = [a_{ij}] = \begin{bmatrix} \log_5 128 & \log_4 5 \\ \log_5 8 & \log_4 25 \end{bmatrix}$. If A_{ij} is the cofactor of a_{ij}, $C_{ij} = \sum_{k=1}^{2} a_{ik}A_{jk}, 1 \le i, j \le 2$, and $C = [C_{ij}]$, then $8|C|$ is equal to :

(A) 288 (B) 262 (C) 222 (D) 242

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Q29. MCQ (Single Correct Answer)

Consider an A. P. of positive integers, whose sum of the first three terms is 54 and the sum of the first twenty terms lies between 1600 and 1800. Then its 11th term is :

(A) 108 (B) 90 (C) 122 (D) 84

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Q30. MCQ (Single Correct Answer)

Let the ellipse $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$ and $E_2 : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1, A < B$ have same eccentricity $\frac{1}{\sqrt{3}}$. Let the product of their lengths of latus rectums be $\frac{32}{\sqrt{3}}$ and the distance between the foci of E_1 be 4. If E_1 and E_2 meet at A, B, C and D, then the area of the quadrilateral ABCD equals :

(A) $\frac{24\sqrt{6}}{5}$ (B) $\frac{18\sqrt{6}}{5}$ (C) $6\sqrt{6}$ (D) $\frac{12\sqrt{6}}{5}$

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Q31. MCQ (Single Correct Answer)

Two parabolas have the same focus (4, 3) and their directrices are the x-axis and the y-axis, respectively. If these parabolas intersect at the points A and B, then $(AB)^2$ is equal to :

(A) 384 (B) 392 (C) 96 (D) 192

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Q32. MCQ (Single Correct Answer)

Let ΔABC be a triangle formed by the lines $7x - 6y + 3 = 0$, $x + 2y - 31 = 0$ and $9x - 2y - 19 = 0$. Let the point (h, k) be the image of the centroid of ΔABC in the line $3x + 6y - 53 = 0$. Then $h^2 + k^2 + hk$ is equal to :

(A) 47 (B) 37 (C) 40 (D) 36

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Q33. MCQ (Single Correct Answer)

The number of solutions of the equation

$\left(\frac{9}{x} - \frac{9}{\sqrt{x}} + 2\right)\left(\frac{2}{x} - \frac{7}{\sqrt{x}} + 3\right) = 0$ is :

(A) 3 (B) 2 (C) 1 (D) 4

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Q34. MCQ (Single Correct Answer)

The least value of **n** for which the number of integral terms in the Binomial expansion of $(\sqrt[3]{7}+\sqrt[12]{11})^n$ is 183, is :

(A) 2184 (B) 2172 (C) 2196 (D) 2148

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Q35. MCQ (Single Correct Answer)

The value of $\lim\limits_{n\to\infty}\left(\sum\limits_{k=1}^{n}\frac{k^3+6k^2+11k+5}{(k+3)!}\right)$ is :

(A) 5/3 (B) 2 (C) 4/3 (D) 7/3

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Q36. MCQ (Single Correct Answer)

Let *y = y(x)* be the solution of the differential equation :

$\cos x(\log_e(\cos x))^2dy+(\sin x-3y\sin x\log_e(\cos x))dx=0$, x ∈ (0, $\frac{\pi}{2}$). If $y(\frac{\pi}{4})=-\frac{1}{\log_e 2}$, then $y(\frac{\pi}{6})$ is equal to :

(A) $\frac{2}{\log_e(3)-\log_e(4)}$ (B) $-\frac{1}{\log_e(4)}$

(C) $\frac{1}{\log_e(4)-\log_e(3)}$ (D) $\frac{1}{\log_e(3)-\log_e(4)}$

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Q37. MCQ (Single Correct Answer)

Let $|z_1-8-2i|\le 1$ and $|z_2-2+6i|\le 2$, $z_1,z_2\in\mathbb{C}$. Then the minimum value of $|z_1-z_2|$ is :

(A) 3 (B) 10 (C) 7 (D) 13

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Q38. MCQ (Single Correct Answer)

Let $x_1,x_2,\ldots,x_{10}$ be ten observations such that $\sum\limits_{i=1}^{10}(x_i-2)=30$, $\sum\limits_{i=1}^{10}(x_i-\beta)^2=98$, $\beta>2$, and their variance is $\frac{4}{5}$. If μ and σ^2 are respectively the mean and the variance of $2(x_1-1)+4\beta, 2(x_2-1)+4\beta,\ldots,2(x_{10}-1)+4\beta$, then $\frac{\beta\mu}{\sigma^2}$ is equal to :

(A) 100 (B) 90 (C) 120 (D) 110

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Q39. MCQ (Single Correct Answer)

Let $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$ and $\vec{b}=2\hat{i}+7\hat{j}+3\hat{k}$. Let $L_1:\vec{r}=(-\hat{i}+2\hat{j}+\hat{k})+\lambda\vec{a},\lambda\in\mathbf{R}$ and $L_2:\vec{r}=(\hat{j}+\hat{k})+\mu\vec{b},\mu\in\mathrm{R}$ be two lines. If the line L_3 passes through the point of intersection of L_1 and L_2 and is parallel to $\vec{a}+\vec{b}$, then L_3 passes through the point :

(A) $(-1,-1,1)$ (B) $(2,8,5)$ (C) $(8,26,12)$ (D) $(5,17,4)$

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Q40. MCQ (Single Correct Answer)

Let the area of the region

$\{(x, y) : 2y \le x^2 + 3,\ y + |x| \le 3,\ y \ge |x - 1|\}$ be A. Then $6A$ is equal to :

(A) 14 (B) 18 (C) 16 (D) 12

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Q41. MCQ (Single Correct Answer)

Let $\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} - 5\hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} = \vec{c} \times \vec{b}$ and $(\vec{a} + \vec{c}) \cdot (\vec{b} + \vec{c}) = 168$. Then the maximum value of $|\vec{c}|^2$ is :

(A) 77 (B) 154 (C) 308 (D) 462

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Q42. MCQ (Single Correct Answer)

Let the line $x + y = 1$ meet the circle $x^2 + y^2 = 4$ at the points A and B. If the line perpendicular to AB and passing through the mid-point of the chord AB intersects the circle at C and D, then the area of the quadrilateral ABCD is equal to :

(A) $\sqrt{14}$ (B) $3\sqrt{7}$ (C) $2\sqrt{14}$ (D) $5\sqrt{7}$

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Q43. MCQ (Single Correct Answer)

Define a relation R on the interval $\left[0, \frac{\pi}{2}\right)$ by x R y if and only if $\sec^2 x - \tan^2 y = 1$. Then R is :

(A) both reflexive and symmetric but not transitive

(B) both reflexive and transitive but not symmetric

(C) reflexive but neither symmetric nor transitive

(D) an equivalence relation

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Q44. MCQ (Single Correct Answer)

Let M and m respectively be the maximum and the minimum values of

$$f(x) = \begin{vmatrix} 1 + \sin^2 x & \cos^2 x & 4\sin 4x \\ \sin^2 x & 1 + \cos^2 x & 4\sin 4x \\ \sin^2 x & \cos^2 x & 1 + 4\sin 4x \end{vmatrix}, x \in R$$

Then $M^4 - m^4$ is equal to :

(A) 1280 (B) 1040 (C) 1215 (D) 1295

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Q45. MCQ (Single Correct Answer)

The integral $80 \int_0^{\frac{\pi}{4}} \left(\frac{\sin\theta + \cos\theta}{9 + 16\sin 2\theta}\right) d\theta$ is equal to :

(A) $3\log 4$ (B) $4\log 3$ (C) $6\log\frac{4}{3}$ (D) $2\log 3$

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Q46. Numerical

Let [t] be the greatest integer less than or equal to t. Then the least value of p ∈ N for which $\lim\limits_{x\to 0^+}\left(x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+\ldots+\left[\frac{p}{x}\right]\right)-x^2\left(\left[\frac{1}{x^2}\right]+\left[\frac{2^2}{x^2}\right]+\ldots+\left[\frac{9^2}{x^2}\right]\right)\right)\geq 1$ is equal to _____.

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Q47. Numerical

Let $S=\left\{m\in\mathbf{Z}:A^{m^2}+A^m=3I-A^{-6}\right\}$, where $A=\begin{bmatrix}2 & -1\\ 1 & 0\end{bmatrix}$. Then $n(S)$ is equal to _____.

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Q48. Numerical

Let S = $\left\{x:\cos^{-1}x=\pi+\sin^{-1}x+\sin^{-1}(2x+1)\right\}$. Then $\sum\limits_{x\in S}(2x-1)^2$ is equal to _____.

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Q49. Numerical

Let $f:(0,\infty)\to\mathbf{R}$ be a twice differentiable function. If for some $a\neq 0, \int\limits_0^1 f(\lambda x)\mathrm{d}\lambda=af(x), f(1)=1$ and $f(16)=\frac{1}{8}$, then $16-f'\left(\frac{1}{16}\right)$ is equal to _____.

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Q50. Numerical

The number of 6-letter words, with or without meaning, that can be formed using the letters of the word MATHS such that any letter that appears in the word must appear at least twice, is _____.

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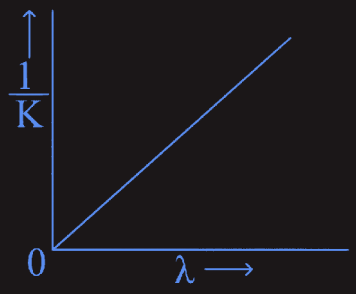
Physics 25 questions

Q51. MCQ (Single Correct Answer)

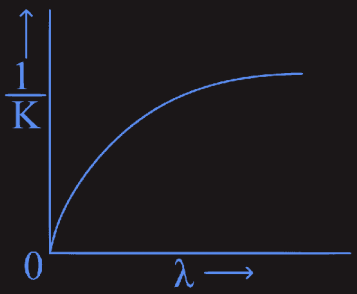
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If λ and K are de Broglie wavelength and kinetic energy, respectively, of a particle with constant mass. The correct graphical representation for the particle will be :

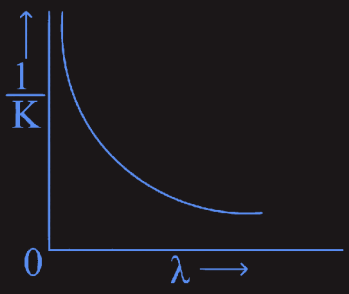
(A)



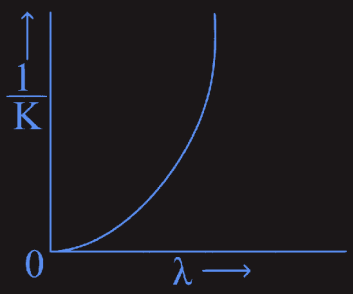
(B)



(C)



(D)



Q52. MCQ (Single Correct Answer)

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Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Electromagnetic waves carry energy but not momentum.

Reason (R) : Mass of a photon is zero.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

(A) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

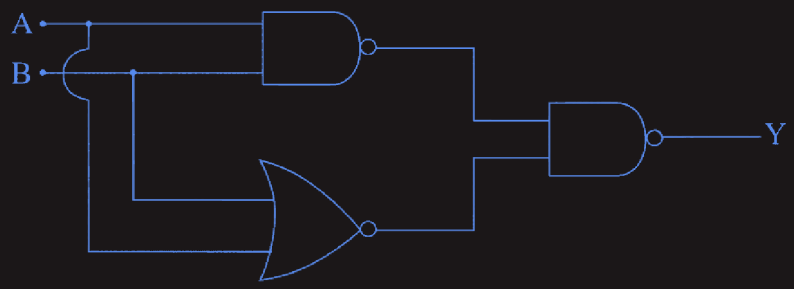
(B) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

(C) **(A)** is false but **(R)** is true

(D) **(A)** is true but **(R)** is false

Q53. MCQ (Single Correct Answer)

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For the circuit shown above, equivalent GATE is :

(A) NOT gate (B) NAND gate (C) AND gate (D) OR gate

Q54. MCQ (Single Correct Answer)

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The pair of physical quantities not having the same dimensions is :

(A) Angular momentum and Planck's constant (B) Torque and energy

(C) Surface tension and impulse (D) Pressure and Young's modulus

---

Q55. MCQ (Single Correct Answer)

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The expression given below shows the variation of velocity (v) with time (t),

$v = \mathrm{At}^2 + \frac{\mathrm{Bt}}{\mathrm{C}+\mathrm{t}}$.

The dimension of ABC is :

(A) $[M^0L^1T^{-2}]$ (B) $[M^0L^2T^{-3}]$ (C) $[M^0L^2T^{-2}]$ (D) $[M^0L^1T^{-3}]$

Q56. MCQ (Single Correct Answer)

The fractional compression $\left(\frac{\Delta V}{V}\right)$ of water at the depth of 2.5 km below the sea level is _____ %. Given, the Bulk modulus of water = 2×10^9 N m^{-2}, density of water = 10^3 kg m^{-3}, acceleration due to gravity $g = 10$ m s^{-2}.

(A) 1.0 (B) 1.25 (C) 1.75 (D) 1.5

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Q57. MCQ (Single Correct Answer)

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Time period of a simple pendulum is longer at the top of a mountain than that at the base of the mountain.

Reason (R) : Time period of a simple pendulum decreases with increasing value of acceleration due to gravity and vice-versa.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**.

(B) **(A)** is true but **(R)** is false.

(C) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(D) **(A)** is false but **(R)** is true.

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Q58. MCQ (Single Correct Answer)

Two projectiles are fired with same initial speed from same point on ground at angles of $(45^\circ - \alpha)$ and $(45^\circ + \alpha)$, respectively, with the horizontal direction. The ratio of their maximum heights attained is :

(A) $\frac{1+\sin\alpha}{1-\sin\alpha}$ (B) $\frac{1+\sin 2\alpha}{1-\sin 2\alpha}$

(C) $\frac{1-\tan\alpha}{1+\tan\alpha}$ (D) $\frac{1-\sin 2\alpha}{1+\sin 2\alpha}$

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Q59. MCQ (Single Correct Answer)

An electric dipole of mass m, charge q, and length l is placed in a uniform electric field $\vec{E} = E_0\hat{i}$. When the dipole is rotated slightly from its equilibrium position and released, the time period of its oscillations will be :

(A) $\frac{1}{2\pi}\sqrt{\frac{ml}{2qE_0}}$ (B) $2\pi\sqrt{\frac{ml}{2qE_0}}$

(C) $2\pi\sqrt{\frac{ml}{qE_0}}$ (D) $\frac{1}{2\pi}\sqrt{\frac{2\,\mathrm{m}l}{\mathrm{q}E_0}}$

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Q60. MCQ (Single Correct Answer)

Consider a long straight wire of a circular cross-section (radius *a*) carrying a steady current *I*. The current is uniformly distributed across this cross-section. The distances from the centre of the wire's cross-section at which the magnetic field [inside the wire, outside the wire] is half of the maximum possible magnetic field, any where due to the wire, will be :

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(A) [*a*/2, 3*a*] (B) [*a*/4, 3*a*/2] (C) [*a*/2, *2a*] (D) [*a*/4, 2*a*]

Q61. MCQ (Single Correct Answer)

A coil of area A and N turns is rotating with angular velocity ω in a uniform magnetic field $\vec{B}$ about an axis perpendicular to $\vec{B}$. Magnetic flux φ and induced emf ε across it, at an instant when $\vec{B}$ is parallel to the plane of coil, are :

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(A) $\varphi = AB, \varepsilon = NAB\omega$ (B) $\varphi = AB, \varepsilon = 0$ (C) $\varphi = 0, \varepsilon = 0$ (D) $\varphi = 0, \varepsilon = NAB\omega$

Q62. MCQ (Single Correct Answer)

Consider I_1 and I_2 are the currents flowing simultaneously in two nearby coils 1 & 2, respectively. If L_1 = self inductance of coil 1, M_{12} = mutual inductance of coil 1 with respect to coil 2, then the value of induced emf in coil 1 will be :

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(A) $e_1 = -L_1\frac{dI_2}{dt} - M_{12}\frac{dI_1}{dt}$ (B) $e_1 = -L_1\frac{dI_1}{dt} + M_{12}\frac{dI_2}{dt}$

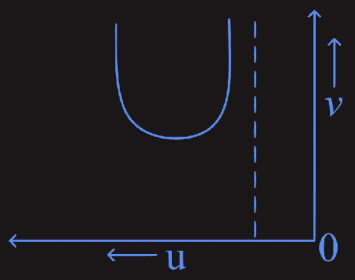
(C) $e_1 = -L_1\frac{dI_1}{dt} - M_{12}\frac{dI_1}{dt}$ (D) $e_1 = -L_1\frac{dI_1}{dt} - M_{12}\frac{dI_2}{dt}$

Q63. MCQ (Single Correct Answer)

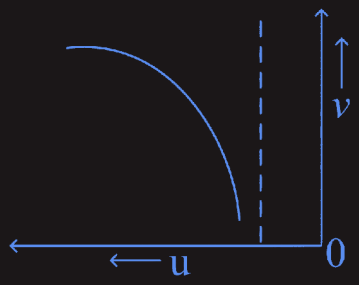
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Let u and v be the distances of the object and the image from a lens of focal length f. The correct graphical representation of u and v for a convex lens when $|u| > f$, is

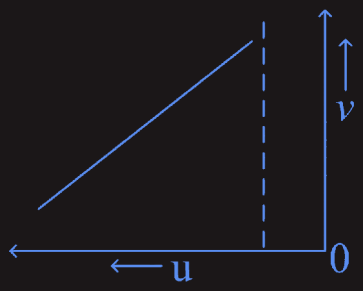
(A)



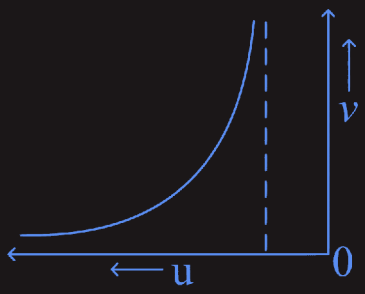
(B)



(C)



(D)



Q64. MCQ (Single Correct Answer)

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The work done in an adiabatic change in an ideal gas depends upon only :

(A) change in its pressure

(B) change in its temperature

(C) change in its specific heat

(D) change in its volume

Q65. MCQ (Single Correct Answer)

A body of mass 'm' connected to a massless and unstretchable string goes in vertical circle of radius 'R' under gravity g. The other end of the string is fixed at the center of circle. If velocity at top of circular path is $n\sqrt{gR}$, where, n ≥ 1, then ratio of kinetic energy of the body at bottom to that at top of the circle is :

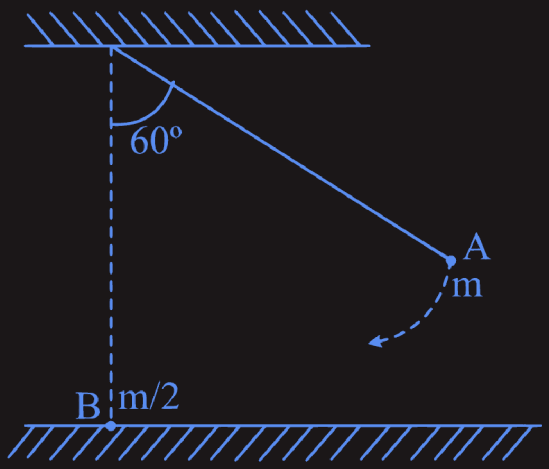
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(A) $\frac{n^2+4}{n^2}$ (B) $\frac{n+4}{n}$ (C) $\frac{n^2}{n^2+4}$ (D) $\frac{n}{n+4}$

Q66. MCQ (Single Correct Answer)

As shown below, bob A of a pendulum having massless string of length 'R' is released from 60° to the vertical. It hits another bob B of half the mass that is at rest on a frictionless table in the center. Assuming elastic collision, the magnitude of the velocity of bob A after the collision will be (take g as acceleration due to gravity.)

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(A) $\frac{4}{3}\sqrt{Rg}$ (B) $\frac{1}{3}\sqrt{Rg}$ (C) $\sqrt{Rg}$ (D) $\frac{2}{3}\sqrt{Rg}$

Q67. Numerical

A container of fixed volume contains a gas at 27°C. To double the pressure of the gas, the temperature of gas should be raised to _____ °C.

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Q68. Numerical

The maximum speed of a boat in still water is 27 km/h. Now this boat is moving downstream in a river flowing at 9 km/h. A man in the boat throws a ball vertically upwards with speed of 10 m/s. Range of the ball as observed by an observer at rest on the bank is _____ cm. (Take $g = 10$ m/s^2)

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Q69. Numerical

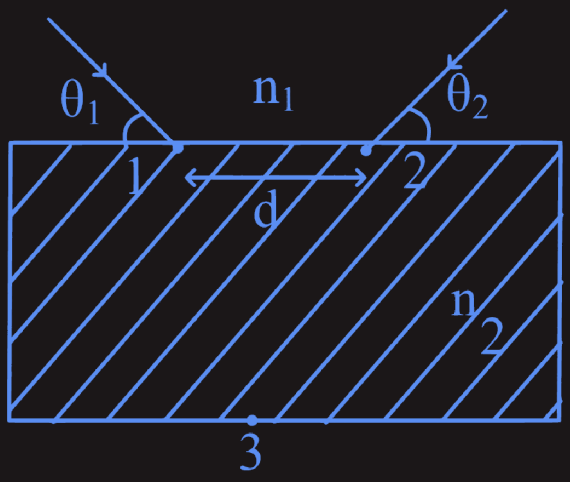
The coordinates of a particle with respect to origin in a given reference frame is (1, 1, 1) meters. If a force of $\vec{F} = \hat{i} - \hat{j} + \hat{k}$ acts on the particle, then the magnitude of torque (with respect to origin) in z-direction is _____.

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Q70. Numerical

Two light beams fall on a transparent material block at point 1 and 2 with angle θ_1 and θ_2, respectively, as shown in figure. After refraction, the beams intersect at point 3 which is exactly on the interface at other end of the block. Given : the distance between 1 and 2, $\mathrm{d} = 4\sqrt{3}\,\mathrm{cm}$ and $\theta_1 = \theta_2 = \cos^{-1}\left(\frac{n_2}{2n_1}\right)$. where refractive index of the block $n_2 >$ refractive index of the outside medium n_1, then the thickness of the block is _____ cm .

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Q71. Numerical

In a hydraulic lift, the surface area of the input piston is 6 cm^2 and that of the output piston is 1500 cm^2. If 100 N force is applied to the input piston to raise the output piston by 20 cm, then the work done is _____ kJ.

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Q72. MCQ (Single Correct Answer)

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Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Emission of electrons in photoelectric effect can be suppressed by applying a sufficiently negative electron potential to the photoemissive substance.

Reason (R) : A negative electric potential, which stops the emission of electrons from the surface of a photoemissive substance, varies linearly with frequency of incident radiation.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is false but (R) is true

(D) (A) is true but (R) is false

Q73. MCQ (Single Correct Answer)

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At the interface between two materials having refractive indices n_1 and n_2, the critical angle for reflection of an em wave is θ_{1C}. The n_2 material is replaced by another material having refractive index n_3 such that the critical angle at the interface between n_1 and n_3 materials is θ_{2C}. If $n_3 > n_2 > n_1$; $\frac{n_2}{n_3} = \frac{2}{5}$ and $\sin\theta_{2C} - \sin\theta_{1C} = \frac{1}{2}$, then θ_{1C} is :

(A) $\sin^{-1}\left(\frac{1}{6n_1}\right)$

(B) $\sin^{-1}\left(\frac{1}{3n_1}\right)$

(C) $\sin^{-1}\left(\frac{5}{6n_1}\right)$

(D) $\sin^{-1}\left(\frac{2}{3n_1}\right)$

Q74. MCQ (Single Correct Answer)

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Match List - I with List - II.

List - I	List - II
(A) Electric field inside (distance r > 0 from center) of a uniformly charged spherical shell with surface charge density σ, and radius R.	(I) σ/ε_0
(B) Electric field at distance r>0 from a uniformly charged infinite plane sheet with surface charge density σ.	(II) $\sigma/2\varepsilon_0$
(C) Electric field outside (distance r>0 from center) of a uniformly charged spherical shell with surface charge density σ, and radius R.	(III) 0
(D) Electric field between 2 oppositely charged infinite plane parallel sheets with uniform surface charge density σ.	(IV) $\frac{\sigma}{\epsilon_0 r^2}$

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(B) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(C) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(D) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

Q75. MCQ (Single Correct Answer)

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Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Choke coil is simply a coil having a large inductance but a small resistance. Choke coils are used with fluorescent mercury-tube fittings. If household electric power is directly connected to a mercury tube, the tube will be damaged.

Reason (R): By using the choke coil, the voltage across the tube is reduced by a factor $\left(R/\sqrt{R^2+\omega^2L^2}\right)$, where ω is frequency of the supply across resistor R and inductor L. If the choke coil were not used, the voltage across the resistor would be the same as the applied voltage.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

(A) (A) is true but (R) is false

(B) Both $(\mathbf{A})$ and $(\mathbf{R})$ are true and $(\mathbf{R})$ is the correct explanation of $(\mathbf{A})$ 3.

(C) Both $(\mathbf{A})$ and $(\mathbf{R})$ are true but $(\mathbf{R})$ is not the correct explanation of $(\mathbf{A})$

(D) (A) is false but (R) is true

Answer Key

Chemistry

1. **D**	2. **D**	3. **D**	4. **A**	5. **B**
6. **B**	7. **D**	8. **C**	9. **A**	10. **C**
11. **C**	12. **D**	13. **B**	14. **C**	15. **D**
16. **C**	17. **A**	18. **C**	19. **A**	20. **B**
21. **160**	22. **341**	23. **16**	24. **13**	25. **15**

Mathematics

26. **A**	27. **C**	28. **D**	29. **B**	30. **A**
31. **D**	32. **B**	33. **D**	34. **A**	35. **A**
36. **D**	37. **C**	38. **A**	39. **C**	40. **A**
41. **C**	42. **C**	43. **D**	44. **A**	45. **B**
46. **24**	47. **2**	48. **5**	49. **112**	50. **1405**

Physics

51. **D**	52. **C**	53. **D**	54. **C**	55. **B**
56. **B**	57. **C**	58. **D**	59. **B**	60. **C**
61. **D**	62. **D**	63. **D**	64. **B**	65. **A**
66. **B**	67. **327**	68. **2000**	69. **2**	70. **6**
71. **5**	72. **B**	73 Not listed*	74. **C**	75. **B**

For numerical answers given as a range, any value within the range is correct.
* A definitive answer is not provided for each question marked Not listed*.

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